



THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ELECTRICAL SYSTEMS ENGINEERING
DEPARTMENT OF INFORMATION & COMMUNICATION ENGINEERING
B. ENG. DEGREE EXAMINATIONS. FIRST SEMESTER 2024/2025 SESSION

COURSE CODE: ICT405 **COURSE TITLE:** Internet Technology and Programming

UNITS: 3.0

TIME ALLOWED: 2½ Hours

Instruction: Read Carefully and Answer any Five (5) questions

Question 1 ✕

- a. Briefly discuss the evolution of video game development from early arcade games to modern AAA titles. (6 marks)
- b. Identify and explain two major technological innovations that significantly influenced game development in the last two decades. (8 marks)
- c. Highlight the difference between the development of console games and PC games in the 1990s. (6 marks)

Question 2 ✓

- a. Compare and contrast the roles of C++ and Python in game development, highlighting strengths and weaknesses. (6 marks)
- b. Name one game engine or framework for each language and describe its core features or use cases. (8 marks)
- c. Why is language performance critical in AAA games, and how does it affect the player experience? (6 marks)

Question 3

- a. Define Object-Oriented Programming and explain how **Encapsulation** and **Inheritance** apply to game development. (6 marks)
- b. Using a game genre of your choice, describe how you would model a class hierarchy (minimum 3 classes). (8 marks)
- c. Explain the benefit of using polymorphism in controlling game entities (e.g., enemies, players, NPCs). (6 marks)

Question 4 ✓

- a. Unity uses a component-based architecture. Explain what this means and how it differs from pure object-oriented design. (6 marks)
- b. Write a simple Unity C# script that moves a player object forward using the `Update()` method. (8 marks)
- c. Briefly explain the purpose of Unity's `Start()` and `FixedUpdate()` methods and how they differ from `Update()`. (6 marks)



THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ELECTRICAL SYSTEMS ENGINEERING
DEPARTMENT OF INFORMATION & COMMUNICATION ENGINEERING
B. ENG. DEGREE EXAMINATIONS. FIRST SEMESTER 2024/2025 SESSION

Question 5

- a. Discuss how **3D game interfaces** differ from 2D interfaces in terms of design considerations and user experience. (6 marks)
- b. Explain the role of the following in a 3D game scene: **Camera, Lighting, and Mesh Renderers**. (8 marks)
- c. Give two examples of good interface design practices in 3D games and why they matter. (6 marks)

Question 6

- a. Explain the importance of **game physics** and **collision detection** in a 3D game environment. (6 marks)
- b. Describe how **Rigidbody** and **Collider** components are used in Unity to implement physical interactions. (8 marks)
- c. Explain the use of **Physics Materials** in Unity and how they affect gameplay. (6 marks)

Question 7

- a. State two advantages and two limitations of using Python for game development. (6 marks)
- b. Choose a Python-based framework (e.g., Pygame) and describe how you would build a simple 2D game idea with it. (8 marks)
- c. In what type of game development scenario would you choose Python over other languages, and why? (6 marks)